



Department of Information Technology

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CCS334 & Big Data Analytics

Name of the Faculty member (s): Mrs.M.Rethina Kumari

Innovative Practice Description

Unit / Topic: Unit 5/HiveQL queries

Course Outcome: CO5

Activity Chosen: Think Aloud Pair Problem Solving

Justification:

The **TAPPS method** encourages students to verbalize their reasoning while solving queries, helping both partners develop deeper conceptual understanding. It promotes collaborative problem-solving, critical thinking, and debugging skills—essential for data analytics tasks using Hive.

Time Allotted for the Activity: 25 minutes

Details of the Implementation:

- Students were paired; one acted as **Thinker**, the other as **Listener**.
- The Thinker explained their reasoning process aloud while solving **Problem 1**.
- The Listener observed, prompted, and gave feedback without taking over the task.
- After completing the first problem, the roles were **switched**.
- The pairs solved **Problem 2** in the reversed roles.
- Finally, each pair discussed insights and shared one key learning with the class.

Question Give:

```
CREATE TABLE sales (  
    order_id INT,  
    customer_name STRING,  
    product STRING,  
    quantity INT,  
    price FLOAT,  
    order_date STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ' ';
```

Problem 1:

Write a HiveQL query to find the total sales amount (quantity × price) for each product.

Switch Roles :

Problem 2:

Write a HiveQL query to display the customer name and total amount spent by each customer who placed more than 3 orders.

CO – PO / PSO mapping:

(1 – Low 2 – Moderate 3 – High)

CO	PO1	PO3	PO4	PO9	PO11	PSO3
CO 5	3	3	2	2	3	3

PO / PSO mapped:

Innovative practice: CO5	PO1	PO3	PO4	PO9	PO11	PSO3
	3	3	2	2	3	3
Justification for correlation	Enhances understanding of data storage and retrieval in Big Data platforms using Hive.	Builds analytical skills for designing and implementing HiveQL queries to solve complex data problems.	Encourages experimentation with query optimization and investigation of efficient data aggregation techniques.	Promotes teamwork through collaborative pair-based problem solving.	Improves analytical communication as students articulate their reasoning processes aloud.	Strengthens ability to apply data analysis techniques to industrial and research problems.

Images / Screenshot of the practice:

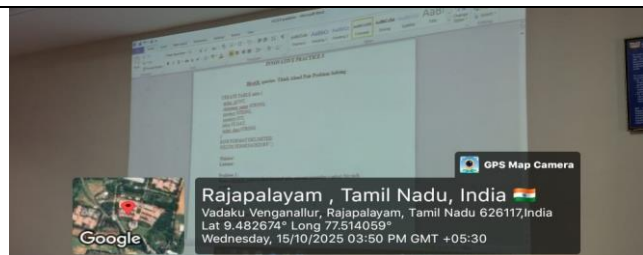


Figure 1 and 2 shows the students preparing for the listed problems

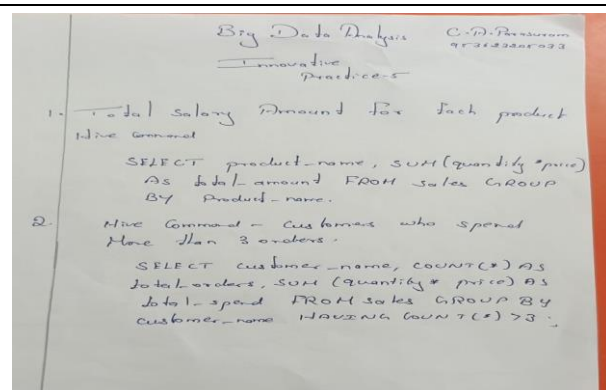
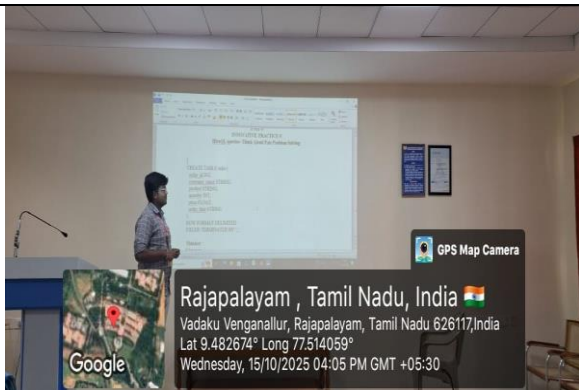


Figure.3 and 4 shows discussion of Student Parasuram and his answers to the problems

Glimpse of Think Pair Share Activity

Feedback of practice from students and other stakeholders:

- Students appreciated the **interactive and engaging nature** of the Think Aloud Pair Problem Solving method.
- It improved their **logical reasoning** and helped identify **syntax and logic errors** in HiveQL queries.
- The role-switching mechanism encouraged **equal participation** and enhanced peer learning.

❖ Benefit of the practice:

- Strengthened **conceptual understanding** of HiveQL and data aggregation.
- Enhanced **communication and collaborative learning** skills.
- Improved **debugging and analytical thinking** abilities.
- Students to **verbalize reasoning**, reinforcing deeper learning.

❖ Challenges faced in implementation:

- Some pairs initially found it difficult to verbalize their reasoning clearly.
- Time management was challenging for complex queries.
- A few students required guidance in forming effective Thinker–Listener pairs.

References:

- <https://hive.apache.org/>
- <https://www.geeksforgeeks.org/hiveql-tutorial/>
- <https://data-flair.training/blogs/hive-ql-queries/>

Signature of Faculty Member

HOD